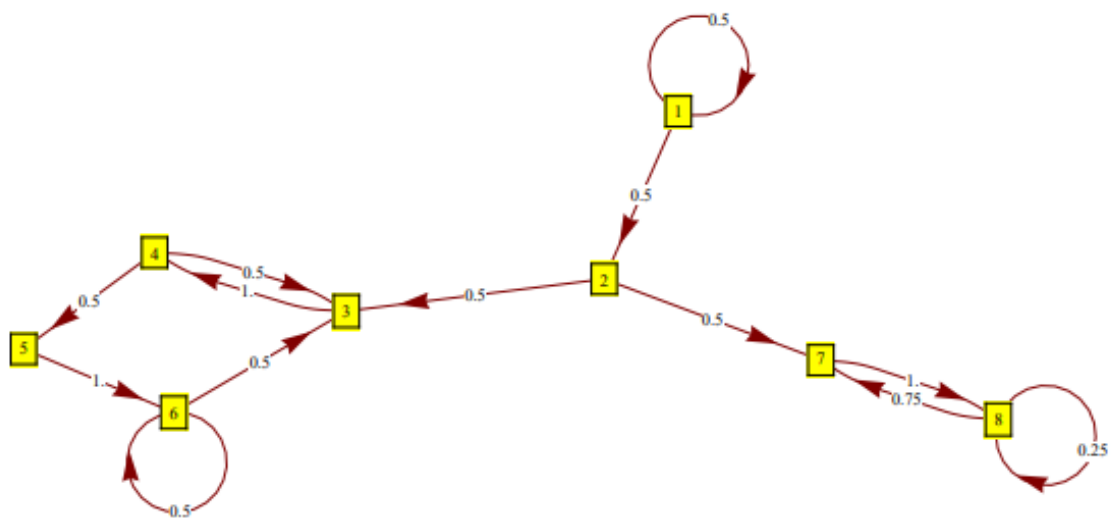


STPA – Assignment 11

1. Consider a Markov Chain whose transition graph is given below

- Identify the classes.
- Find transient and recurrent states.
- Find periods of all states
- Create TPM



Specify the classes of the following Markov chains.

$$\mathbb{P}_1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.3 & 0.6 & 0.1 & 0 \\ 0 & 0 & 0.2 & 0.8 \\ 0 & 0 & 0.9 & 0.1 \end{pmatrix} \end{matrix} \quad \mathbb{P}_2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

2.

Find the communicating classes associated with the stochastic matrix

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1/3 & 0 & 0 & 1/3 & 1/3 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

- 3.
4. A mouse is in a 2×3 maze, form a room, the mouse can go any available room which is either up, or down or to the right or to the left at random. Room 6 is full of cheese and there is a cat in room 3. If the mouse gets to either room 3 or 6, stays there forever. Find the probability that the mouse end up in room 6 assuming that it starts at room i.

1	2	3 CAT
4	5	6 CHEESE